Unit: 6 Unit 7: Factoring
Topic: Sum and Difference of Cubes

Name: ____________
Date: ________

Quiz (Carolina Reaper)

Q1. A shipping company has $x^3 + 8$ cubic meters of cargo to transport. Which factored form is equivalent?
(A) $(x + 2)(x^2 - 2x + 4)$
(B) $(x - 2)(x^2 + 2x + 4)$
(C) $x(x^2 + 8)$
(D) $(x + 2)(x^2 - 2x - 4)$
(E) $(x - 2)(x^2 + 2x - 4)$

Q2. The traffic flow rate is modeled by $x^3 - 27$. Which factored form is equivalent?
(A) $(x - 3)(x^2 + 3x + 9)$
(B) $(x + 3)(x^2 - 3x + 9)$
(C) $x(x^2 - 27)$
(D) $(x - 3)(x^2 - 3x - 9)$
(E) $(x + 3)(x^2 + 3x - 9)$

Q3. A logistics company has $x^3 - 64$ cubic meters of storage space. Which factored form is equivalent?
(A) $(x - 4)(x^2 + 4x + 16)$
(B) $(x + 4)(x^2 - 4x + 16)$
(C) $x(x^2 - 64)$
(D) $(x - 4)(x^2 - 4x - 16)$
(E) $(x + 4)(x^2 + 4x - 16)$

Q4. The delivery time is modeled by $x^3 + 125$. Which factored form is equivalent?
(A) $(x + 5)(x^2 - 5x + 25)$
(B) $(x - 5)(x^2 + 5x + 25)$
(C) $x(x^2 + 125)$
(D) $(x + 5)(x^2 + 5x - 25)$
(E) $(x - 5)(x^2 - 5x - 25)$

Q5. A transportation company has $x^3 - 1000$ cubic meters of cargo to transport. Which factored form is equivalent?
(A) $(x - 10)(x^2 + 10x + 100)$
(B) $(x + 10)(x^2 - 10x + 100)$
(C) $x(x^2 - 1000)$
(D) $(x - 10)(x^2 - 10x - 100)$
(E) $(x + 10)(x^2 + 10x - 100)$

Q6. The traffic flow rate is modeled by $x^3 + 216$. Which factored form is equivalent?
(A) $(x + 6)(x^2 - 6x + 36)$
(B) $(x - 6)(x^2 + 6x + 36)$
(C) $x(x^2 + 216)$
(D) $(x + 6)(x^2 + 6x - 36)$
(E) $(x - 6)(x^2 - 6x - 36)$

Q7. A logistics company has $x^3 - 343$ cubic meters of storage space. Which factored form is equivalent?
(A) $(x - 7)(x^2 + 7x + 49)$
(B) $(x + 7)(x^2 - 7x + 49)$
(C) $x(x^2 - 343)$
(D) $(x - 7)(x^2 - 7x - 49)$
(E) $(x + 7)(x^2 + 7x - 49)$

Q8. The delivery time is modeled by $x^3 + 1000$. Which factored form is equivalent?
(A) $(x + 10)(x^2 - 10x + 100)$
(B) $(x - 10)(x^2 + 10x + 100)$
(C) $x(x^2 + 1000)$
(D) $(x + 10)(x^2 + 10x - 100)$
(E) $(x - 10)(x^2 - 10x - 100)$

Open Ended Questions (FRQ)

Q9. Factor the expression $x^3 - 512$ and explain the reasoning behind your answer in 5 lines or less.

Q10. Factor the expression $x^3 + 729$ and explain the reasoning behind your answer in 5 lines or less.

Chef's Tips

- Tip 1: Ensure students understand the difference between sum and difference of cubes.
- Tip 2: Remind students to check their answers by multiplying the factors.
- Tip 3: Encourage students to use real-world examples, such as transportation and logistics, to illustrate the concepts.